Chapter 8. Bootstrap Confidence Intervals

Normal, standard, bootstrap-t, percentile and BCa intervals

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Overview

Of the methods using bootstrap resampling, the most frequently used and most developed is the estimation of confidence intervals. The best-known way to obtain a confidence interval for a parameter θ uses an estimator $\hat{\theta}_n$ of θ. A bootstrap confidence interval uses the bootstrap sampling distribution of the estimator $\hat{\theta}_n$ of θ to obtain a confidence interval for θ. This chapter explains them with the following example.

Example 8.1 (Team Fortress 2 data)

Figure 8.1 of the book is a histogram of the Team Fortress 2 play time X_i of $n = 2322$ randomly chosen gamers who bought "Team Fortress 2" on the game platform Steam (`TeamFortress2.csv`). To estimate the mean μ of all Steam users who enjoy Team Fortress 2, the sample mean $\bar{X}_n$ is computed and used to obtain a confidence interval for μ.

When $X_1, \ldots, X_n$ are a random sample from a population F, the sample mean $\bar{X}_n$ has the following properties. With population mean μ and variance σ^2,

$$E(\bar{X}_n) = \mu, \qquad \mathrm{Var}(\bar{X}_n) = \frac{\sigma^2}{n} \approx \frac{s^2}{n},$$

where s^2 is the sample variance. And by the central limit theorem,

$$\frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}} \approx \frac{\bar{X}_n - \mu}{s/\sqrt{n}} \dot{\sim} N(0,1).$$

So the 95% interval by the normal approximation is $\mu \in \bar{X}_n \pm z_{0.025} s/\sqrt{n}$, with asymptotically approximately 95% confidence, or coverage probability 0.95: $P(\mu \in \bar{X}_n \pm z_{0.025} s/\sqrt{n}) \approx 0.95$. But when the population is very skewed, as in Example 8.1, convergence to normality by the CLT is very slow, so the normal-approximation interval cannot be guaranteed to have coverage close to 95% at the given sample size n.

Without using the asymptotic distribution of $\bar{X}_n$, estimating its sampling distribution directly by bootstrap resampling gives a more accurate interval. More generally: when the estimator of a parameter $\theta = \theta(F)$ of the population F is $\hat{\theta}_n = T(\mathbb{X}_n)$, we look at how to obtain a $1-\alpha$ interval for θ from the bootstrap sampling distribution of $\hat{\theta}_n$ (or the B bootstrap replicates $\{\hat{\theta}_n^{*b} : b = 1, \ldots, B\}$) and the bootstrap estimate $\widehat{\mathrm{se}}_{\mathrm{Boot}}$ of the standard error. Unlike the bootstrap estimate $\widehat{\mathrm{se}}_{\mathrm{Boot}}$ of the standard error, of which there was only one, there are several methods of bootstrap interval estimation. We look at six representative bootstrap intervals:

- the normal-approximation bootstrap interval;
- the basic bootstrap interval;
- the bootstrap-t interval, or studentized pivotal interval;
- the bootstrap percentile interval, or quantile interval;
  - the bias-corrected (BC) percentile interval;
  - the bias-corrected and accelerated (BCa) percentile interval.

There are more interval methods,[1] but the bootstrap-t and "BCa" methods are the most accurate, and percentile methods are very easy to implement generally, so knowing these methods suffices for applications.

[1] See Davison and Hinkley (1997) and Carpenter and Bithell (2000).

8.1 Normal-approximation bootstrap intervals

First the simplest, the normal-approximation bootstrap interval. In many situations the standardized estimator $\hat{\theta}_n$ is approximately normal: $(\hat{\theta}_n - \theta)/\operatorname{se}(\hat{\theta}_n) \dot{\sim} N(0,1)$. If the estimator $\hat{\theta}_n$ has a non-negligible bias $\operatorname{bias}(\hat{\theta}_n) = E(\hat{\theta}_n) - \theta$, it must be adjusted:

$$\frac{\hat{\theta}_n - \operatorname{bias}(\hat{\theta}_n) - \theta}{\operatorname{se}(\hat{\theta}_n)} \dot{\sim} N(0,1). \tag{8.1}$$

When this can be accepted (when the normal approximation can be assumed accurate), the asymptotically approximate $1-\alpha$ interval is $\theta \in \hat{\theta}_n - \operatorname{bias}(\hat{\theta}_n) \pm z_{\alpha/2} \operatorname{se}(\hat{\theta}_n)$. The standard error $\operatorname{se}(\hat{\theta}_n)$ and bias $\operatorname{bias}(\hat{\theta}_n)$ are unknown, so substituting the bootstrap estimates $\widehat{\operatorname{se}}_{\text{Boot}}$ and $\widehat{\operatorname{bias}}_{\text{Boot}}$,

$$\theta \in \hat{\theta}_n - \widehat{\operatorname{bias}}_{\text{Boot}}(\hat{\theta}_n) \pm z_{\alpha/2} \widehat{\operatorname{se}}_{\text{Boot}}(\hat{\theta}_n),$$

is the **normal-approximation bootstrap interval**.

Note that this method uses only two values of the bootstrap resampling results, $\widehat{\operatorname{se}}_{\text{Boot}}$ and $\widehat{\operatorname{bias}}_{\text{Boot}}$: of the bootstrap sampling distribution, which potentially contains a great deal of information, it uses only the mean and variance. If the shape of the bootstrap sampling distribution of $\hat{\theta}_n$ is similar to a normal, all the remaining information is contained in the normal approximation, and the normal-approximation bootstrap interval is asymptotically exact. Conversely, if the bootstrap sampling distribution does not resemble a normal, there is no guarantee that this interval is exact.

Example 8.2

For the Team Fortress 2 data of Example 8.1, the sample mean is $\bar{X}_n = 70.64$, and its bootstrap sampling distribution is in Figure 8.2 of the book. The standard error estimate is $\widehat{\operatorname{se}}_{\text{Boot}} = 5$ and the bias estimate $\widehat{\operatorname{bias}}_{\text{Boot}} = 70.56 - 70.64 = -0.08$, so the 95% normal-approximation bootstrap interval for μ is

$$\mu \in \bar{X}_n - \widehat{\operatorname{bias}}_{\text{Boot}}(\bar{X}_n) \pm z_{\alpha/2} \widehat{\operatorname{se}}_{\text{Boot}}(\bar{X}_n) = 70.64 + 0.08 \pm 1.96 \times 5 = (60.9, 80.5).$$

In R:

```
> TFtime = read.csv("data/TeamFortress2.csv")$x
> (hat.theta = mean(TFtime))
[1] 70.64311
> n = length(TFtime)
> B = 1999
> set.seed(1)
> theta.boot = replicate(B, mean(sample(TFtime, n, replace =
    TRUE)))
> (se.est = sd(theta.boot))
[1] 5.003227
> (bias.est = mean(theta.boot) - hat.theta)
[1] -0.07757111
> (norm.app.boot.CI = hat.theta - bias.est + se.est * qnorm
    (0.025) *c(1,-1))
[1] 60.91453 80.52683
```

As Figure 8.2 shows, the bootstrap sampling distribution is approximately normal in shape, so this interval can be trusted.

The normal-approximation bootstrap interval is exact only when the normal approximation (8.1) holds. If the sample is small and the normal approximation of the standardized statistic does not occur, or the asymptotic distribution of the statistic is not normal even for large samples, this interval cannot be used.

8.2 The basic bootstrap interval

Rewriting the assumption (8.1) behind the normal-approximation interval gives $\hat{\theta}_n - \theta \dot{\sim} N(\text{bias}, \text{se}^2)$. Here, instead of restricting the distribution of $\hat{\theta}_n - \theta$ to be normal, suppose freely that it has some (continuous) distribution H:

$$\hat{\theta}_n - \theta \sim H. \tag{8.2}$$

This distribution H has mean bias and variance se^2, but otherwise its shape is unspecified: a nonparametric model. With the quantile function $Q_H(p) = H^{-1}(p)$ of H,

$$\begin{aligned} 1-\alpha &= P(Q_H(\tfrac{\alpha}{2}) \le \hat{\theta}_n - \theta \le Q_H(1-\tfrac{\alpha}{2})) \\ &= P(\hat{\theta}_n - Q_H(1-\tfrac{\alpha}{2}) \le \theta \le \hat{\theta}_n - Q_H(\tfrac{\alpha}{2})). \end{aligned} \tag{8.3}$$

If this distribution H does not depend on θ, i.e. if $\hat{\theta}_n - \theta$ is a **pivot**,[2] the probability (8.3) holds for every θ, and so $\hat{\theta}_n - Q_H(1-\frac{\alpha}{2}) \le \theta \le \hat{\theta}_n - Q_H(\frac{\alpha}{2})$ is a $1-\alpha$ interval for θ.

Call the bootstrap-resampled $\hat{\theta}_n$ the bootstrap replicate $\hat{\theta}_n^*$; the distribution of $\hat{\theta}_n^*$ is the bootstrap sampling distribution G^*. The **basic bootstrap interval** (basic bootstrap C.I.) is the interval approximating the distribution H (8.2) of $\hat{\theta}_n - \theta$ by the distribution of $\hat{\theta}_n^* - \hat{\theta}_n$ (call it H^*):

$$\hat{\theta}_n - Q_{H^*}(1-\tfrac{\alpha}{2}) \le \theta \le \hat{\theta}_n - Q_{H^*}(\tfrac{\alpha}{2}). \tag{8.4}$$

The distribution H^* is the distribution G^* of $\hat{\theta}_n^*$ shifted left by $\hat{\theta}_n$, i.e. for $p \in (0,1)$

$$Q_{H^*}(p) = Q_{G^*}(p) - \hat{\theta}_n,$$

so (8.4) can be rewritten

$$2\hat{\theta}_n - Q_{G^*}(1-\tfrac{\alpha}{2}) \le \theta \le 2\hat{\theta}_n - Q_{G^*}(\tfrac{\alpha}{2}). \tag{8.5}$$

A worked example of the basic bootstrap interval is given in Example 8.3, after the bootstrap-t interval.

The basic bootstrap interval relies on the bootstrap approximation that for all θ

$$H^* \sim \hat{\theta}_n^* - \hat{\theta}_n \overset{d}{\approx} \hat{\theta}_n - \theta \sim H. \tag{8.6}$$

This method makes no specific assumption about the distribution of $\hat{\theta}_n$ and can be trusted, but it requires the questionable assumption (8.2) that $\hat{\theta}_n - \theta$ is a pivot. If this assumption fails, the convergence of H^* to H[3] is slow, so the accuracy of the basic bootstrap interval may be low.

8.3 The bootstrap-t interval

The bootstrap-t interval (bootstrap-t C.I.) is also called the studentized pivotal interval. Even when the distribution H of $\hat{\theta}_n - \theta$ and the distribution H^* of $\hat{\theta}_n^* - \hat{\theta}_n$ are not approximated as in (8.6), the following bootstrap approximation using the studentized pivot can be more accurate:

$$H^* \sim \frac{\hat{\theta}_n^* - \hat{\theta}_n}{\widehat{\text{se}}^*} \overset{d}{\approx} \frac{\hat{\theta}_n - \theta}{\widehat{\text{se}}} \sim H. \tag{8.7}$$

If the population is $N(\theta, \sigma^2)$ and $\hat{\theta}_n = \bar{x}_n$, then $(\hat{\theta}_n - \theta)/\widehat{\text{se}} = \sqrt{n}(\bar{x}_n - \theta)/s \sim t_{n-1}$, so here the distribution H is the t-distribution with $n-1$ degrees of freedom. Without normality, H cannot

[2] For a parameter θ of a family $\mathcal{F}$ and statistics T_1, T_2, a random variable $S = S(T_1, T_2, \theta)$ whose distribution is the same for every population $F \in \mathcal{F}$ is called a pivot (for θ). If S is a pivot, its distribution is the same for every population in the family, so it does not depend on the parameter θ nor on the other parameters η of the family.

[3] Convergence here means that as the sample size n grows, the distribution H^* of $\hat{\theta}_n^* - \hat{\theta}_n$ converges to the distribution H of $\hat{\theta}_n - \theta$.

be assumed to be a t-distribution, so the key idea is to approximate H by H^* using the bootstrap. (The name "bootstrap-t" comes from this.)

On the right side of (8.7), $\widehat{\text{se}} = \widehat{\text{se}}(\hat{\theta}_n) \approx \sqrt{\text{Var}(\hat{\theta}_n \mid F)}$ is the standard error estimate of $\hat{\theta}_n = T(\mathbb{X}_n)$ computed from $\mathbb{X}_n$ sampled from the population F. This estimate is given by the given sample $\mathbb{X}_n = \{x_1, \ldots, x_n\}$ (or the empirical distribution F_n). E.g. with the bootstrap standard error estimate,

$$\widehat{\text{se}} = \widehat{\text{se}}_{\text{Boot}}. \tag{8.8}$$

When the standard error is easy to compute, e.g. if $\hat{\theta}_n = \bar{x}_n$,

$$\widehat{\text{se}} = s/\sqrt{n} = \sqrt{\frac{\sum_{i=1}^n (x_i - \bar{x}_n)^2}{n(n-1)}}. \tag{8.9}$$

With a parametric model, the maximum likelihood estimate of the population standard deviation may be used instead of the sample standard deviation s. Either way, $\widehat{\text{se}}$ is determined by the given sample.

Similarly, on the left of (8.7), $\widehat{\text{se}}^* = \widehat{\text{se}}^*(\hat{\theta}_n^*) \approx \sqrt{\text{Var}(\hat{\theta}_n^* \mid F_n)}$ is the standard error estimate of $\hat{\theta}_n^* = T(\mathbb{X}_n^*)$ estimated from the bootstrap sample $\mathbb{X}_n^* = \{x_1^*, \ldots, x_n^*\}$ resampled from F_n (the empirical distribution). Note that it is computed from the bootstrap sample $\mathbb{X}_n^*$, not the original sample $\mathbb{X}_n$: the $\widehat{\text{se}}$ computed from the original sample and the $\widehat{\text{se}}^*$ computed from the bootstrap sample differ. When the standard error estimate is easy to compute, e.g. if $\hat{\theta}_n = \bar{x}_n$ then $\hat{\theta}_n^* = \bar{x}_n^*$ and $\widehat{\text{se}}^* = s^*/\sqrt{n} = \sqrt{\frac{\sum_{i=1}^n (x_i^* - \bar{x}_n^*)^2}{n(n-1)}}$.

Otherwise (when $\widehat{\text{Var}}(\hat{\theta}_n)$ is hard to express by a formula), $\hat{\theta}_n^* = T(\mathbb{X}_n^*)$ is bootstrap-resampled to compute $\widehat{\text{se}}^*_{\text{Boot}}$. This bootstrap resampling resamples again from the bootstrap-resampled $\mathbb{X}_n^*$, so it is called **second-level bootstrap resampling**. Figure 8.3 of the book shows schematically the first-level resampling done B times and the second-level resampling done C times.

The studentized pivotal interval using the bootstrap approximation (8.7) is

$$\hat{\theta}_n - \widehat{\text{se}} \times Q_{H^*}(1 - \tfrac{\alpha}{2}) \le \theta \le \hat{\theta}_n - \widehat{\text{se}} \times Q_{H^*}(\tfrac{\alpha}{2}), \tag{8.10}$$

where $\widehat{\text{se}}$ is given by (8.8) or (8.9), and $Q_{H^*}(p)$ is the p-quantile (the $p \times B$-th smallest value) of the bootstrap-resampled studentized pivots

$$\frac{\hat{\theta}_n^{*(1)} - \hat{\theta}_n}{\widehat{\text{se}}^{*(1)}}, \ \frac{\hat{\theta}_n^{*(2)} - \hat{\theta}_n}{\widehat{\text{se}}^{*(2)}}, \ \ldots, \ \frac{\hat{\theta}_n^{*(B)} - \hat{\theta}_n}{\widehat{\text{se}}^{*(B)}}$$

when bootstrap resampling is repeated B times. With $B = 2000$ bootstrap resamples and second-level resampling C times to approximate $\widehat{\text{se}}^*$, a total of $B \times C$ bootstrap resamples is needed, so a large C takes very long computation. Second-level resampling of about $C = 25$ is recommended. For intervals, $B = 1999$ or more bootstrap resamples are recommended.

The studentized pivotal interval (8.10) is more accurate the better the bootstrap approximation (8.7) holds. Its coverage probability $P(\theta \in \text{CI})$ is asymptotically $1 - \alpha$:

$$\begin{aligned}
P(\theta \in \text{CI}) &= P\big(\hat{\theta}_n - \widehat{\text{se}} \times Q_{H^*}(1 - \tfrac{\alpha}{2}) \le \theta \le \hat{\theta}_n - \widehat{\text{se}} \times Q_{H^*}(\tfrac{\alpha}{2})\big) \\
&\approx P\big(\hat{\theta}_n - \widehat{\text{se}} \times Q_H(1 - \tfrac{\alpha}{2}) \le \theta \le \hat{\theta}_n - \widehat{\text{se}} \times Q_H(\tfrac{\alpha}{2})\big) \\
&= P\Big(Q_H(\tfrac{\alpha}{2}) \le \frac{\hat{\theta}_n - \theta}{\widehat{\text{se}}} \le Q_H(1 - \tfrac{\alpha}{2})\Big) \\
&= 1 - \alpha.
\end{aligned}$$

Example 8.3 (Bootstrap-t and basic bootstrap intervals)

For the Team Fortress 2 data, obtain the basic bootstrap interval and the bootstrap-t interval for the population mean θ. We use the nonparametric bootstrap resampling example. Since the statistic is $\hat{\theta}_n = \bar{x}_n$, $\widehat{\text{se}}$ and $\widehat{\text{se}}^*$ can be computed by (8.9) without assuming a parametric model for the population. For each bootstrap sample $\mathbb{X}_n^{*(b)} = \{x_1^{*(b)}, \ldots, x_n^{*(b)}\}$ compute

$$\hat{\theta}_n^{*(b)} = \bar{x}_n^{*(b)}, \qquad \widehat{\text{se}}^{*(b)} = \sqrt{\frac{\sum_{i=1}^n (x_i^{*(b)} - \bar{x}_n^{*(b)})^2}{n(n-1)}}, \qquad b = 1, \ldots, B, \tag{8.11}$$

and from them the studentized pivots

$$t^{*(b)} = \frac{\hat{\theta}_n^{*(b)} - \hat{\theta}_n}{\widehat{\text{se}}^{*(b)}}, \qquad b = 1, \ldots, B.$$

Sorting these $t^{*(b)}$ to order statistics $t^*_{(1)} \le t^*_{(2)} \le \cdots \le t^*_{(B)}$, $Q_{H^*}(p) = t^*_{(\lfloor p(B+1) \rfloor)}$.[a] With $B = 1999$ and $\alpha = 0.05$, $Q_{H^*}(\alpha/2) = t^*_{(50)}$ and $Q_{H^*}(1-\alpha/2) = t^*_{(1950)}$: the 50th and 1950th smallest $t^{*(b)}$. The R code below carries out the bootstrap resampling.

```
> B = 1999
> boot.out = replicate(B, {
+             TFtime.b = sample(TFtime, n, replace = TRUE)
+             c(mean(TFtime.b), sd(TFtime.b)/sqrt(n))
+   }) %>% t()
> colnames(boot.out) = c("theta*","se*")
> head(boot.out)
       theta*      se*
[1,] 63.77382 4.428347
[2,] 64.20879 4.390289
[3,] 65.45922 4.332700
[4,] 69.65388 4.917695
[5,] 74.98221 5.243905
[6,] 77.79625 5.621729
> hat.theta = mean(TFtime)
> t.pivot = sort( (boot.out[,"theta*"] - hat.theta) / boot.out[,
    "se*"] )
```

By (8.10) the 95% bootstrap-t interval for θ follows; here $\widehat{\text{se}}$ is not the bootstrap estimate of the standard error but the estimate by formula (8.11).

```
> se.hat.theta = sd(TFtime) / sqrt(n)
> hat.theta - se.hat.theta * c(t.pivot[1950], t.pivot[50])
[1] 61.82523 82.17272
```

The quantiles can also be obtained with `quantile()`:

```
> hat.theta - se.hat.theta * quantile(t.pivot, probs= c
    (1-0.025,0.025),type = 1)
   97.5%     2.5%
61.82523 82.17272
```

By (8.5) the basic bootstrap interval is:

```
> boot.dist = sort(boot.out[,"theta*"])
> 2*hat.theta - c(boot.dist[1950], boot.dist[50])
[1] 60.78312 80.57468
```

[a] The bootstrap-approximated sampling distribution H^* or G^* is a discrete distribution with mostly B values. For a discrete distribution the quantile functions Q_{H^*} or Q_{G^*} can be defined exactly, but in practice, considering that this discrete distribution approximates a continuous (bootstrap) distribution, various

approximations of the quantile function are used. E.g. R's `quantile()`, which computes quantiles of given data, offers 9 different kinds of quantile computation. Its argument `type = 1` computes the quantile as defined here; the default is `type = 7`.

Example 8.4 (Second-level bootstrap)

When the standard error estimate $\widehat{\text{se}}$ of the statistic $\hat{\theta}$ cannot be given by a formula, a second-level bootstrap is needed. With $C = 25$ repetitions, the second-level bootstrap is carried out additionally for each bootstrap sample as below. (It takes a very long time.)

```
> B = 1999
> C = 25
> boot.out = replicate(B, {
+   TFtime.b = sample(TFtime, n, replace = TRUE)
+   theta.boot = mean(TFtime.b)
+   theta.boot.se = sd( replicate(C, mean (sample(TFtime.b, n,
    replace = TRUE)) ))
+   c(theta.boot, theta.boot.se)
+ }) %>% t()
> colnames(boot.out) = c("theta*","se*")
> head(boot.out)
       theta*      se*
[1,] 70.63376 4.071472
[2,] 56.81318 6.023104
[3,] 74.89871 5.378420
[4,] 65.11792 4.644929
[5,] 74.22063 5.994275
[6,] 67.25078 4.730485
> hat.theta = mean(TFtime)
> t.pivot = sort( (boot.out[,"theta*"] - hat.theta) / boot.out[,
    "se*"] )
```

By (8.10) the 95% bootstrap-t interval for θ follows, here with the bootstrap estimate of the standard error as $\widehat{\text{se}}$.

```
> se.hat.theta.boot = sd(boot.out[,"theta*"])
> hat.theta - se.hat.theta * c(t.pivot[1950], t.pivot[50]) #
    bootstrap-t
[1] 61.11097 81.99433
```

The bootstrap-t interval performs best when the studentized statistic $t := (\hat{\theta}_n - \theta)/\widehat{\text{se}}$ is a pivot. If the pivot of the statistic $\hat{\theta}_n$ and the parameter θ has another form, the resulting interval is necessarily less accurate. E.g. if the population is $N(\mu, \sigma^2)$ and $\theta = \sigma^2$, the pivot of its estimator $\hat{\theta}_n = S^2$ is

$$(n-1)\frac{S^2}{\sigma^2} \sim \chi^2_{n-1},$$

so $(S^2 - \sigma^2)/\widehat{\text{se}}$ cannot be said to be a pivot. Then a transformation of $\hat{\theta}_n$ can improve the accuracy of the basic or bootstrap-t interval. E.g. if $(\log(\hat{\theta}_n) - \log(\theta))/\widehat{\text{se}}\{\log(\hat{\theta}_n)\}$ is approximately a pivot, the bootstrap-t interval for the parameter $\log(\theta)$,

$$L \le \log(\theta) \le U,$$

transforms to the interval for θ

$$\exp(L) \le \theta \le \exp(U).$$

(Supplied: since exp is increasing, $\{L \le \log\theta \le U\} = \{e^L \le \theta \le e^U\}$, so the coverage probability is unchanged.)

8.4 The bootstrap percentile interval

The bootstrap percentile (quantile) interval (percentile C.I.) is very simple to compute. Given an estimator $\hat{\theta}_n$ of θ and its bootstrap replicates $\{\hat{\theta}_n^{*(b)} : b = 1, \ldots, B\}$ (or the bootstrap sampling distribution G^* of $\hat{\theta}_n$), the bootstrap percentile interval at level $1-\alpha$ is given by the $\frac{\alpha}{2}$- and $(1-\frac{\alpha}{2})$-quantiles of G^*:

$$\theta \in (Q_{G^*}(\tfrac{\alpha}{2}),\ Q_{G^*}(1-\tfrac{\alpha}{2})). \tag{8.12}$$

With $B = 1999$ and $\alpha = 0.05$, the lower and upper limits are the 50th and 1950th of the ordered bootstrap replicates. Writing the sorted $\hat{\theta}_n^{*(b)}$ as order statistics $\hat{\theta}^*_{(1)} \le \cdots \le \hat{\theta}^*_{(B)}$,

$$\theta \in (\hat{\theta}^*_{(50)},\ \hat{\theta}^*_{(1950)})$$

is the 95% percentile interval.

Example 8.5

The sampling distribution G^* obtained by $B = 1999$ bootstrap resamples of $\hat{\theta}_n = \bar{X}_n$ for the Team Fortress 2 data is shown in Figure 8.4 of the book. The 95% percentile interval for the population mean θ is $(\hat{\theta}^*_{(50)}, \hat{\theta}^*_{(1950)}) = (60.7, 80.5)$, computed as follows:

```
> set.seed(1)
> sort.b = replicate(B, mean(sample(TFtime, n, replace = TRUE)))
    %>% sort()
> (perc.int = c(sort.b[50],sort.b[1950]))
[1] 60.71154 80.50310
```

This percentile interval is marked in Figure 8.4.

The 95% percentile interval is the "middle" 95% of the bootstrap sampling distribution, so it can easily be computed by any bootstrap resampling. Why can this interval be trusted? Inferring the location of the parameter θ from the sampling distribution of a statistic may look natural, but this understanding, as the statistician Peter Hall said, amounts to "looking up the wrong statistical tables backwards" (Hall, 2013, p. 36): it finds the approximate answer only by accident.

The percentile interval rests on the following assumption. When introducing the basic and bootstrap-t intervals, we saw that one may transform to make $(\hat{\theta}_n - \theta)/\widehat{\text{se}}$ a pivot, e.g. the log transform $(\hat{\theta}_n \mapsto \varphi(\hat{\theta}_n) := \log(\hat{\theta}_n))$. But in a given problem such a transform can be hard to find. The percentile interval assumes that there exists some monotone increasing transform φ such that the transformed values form a pivot following the standard normal distribution, i.e.

$$\varphi(\hat{\theta}_n) - \varphi(\theta) \sim N(0,1). \tag{8.13}$$

(Or approximately a pivot.) Transforming the bootstrap replicates $\hat{\theta}_n^*$ too,

$$\varphi(\hat{\theta}_n^*) - \varphi(\hat{\theta}_n) \overset{d}{\approx} \varphi(\hat{\theta}_n) - \varphi(\theta) \sim N(0,1), \tag{8.14}$$

so the distribution of $\varphi(\hat{\theta}_n^*) - \varphi(\hat{\theta}_n)$ is also approximately standard normal. The left side $\varphi(\hat{\theta}_n^*) - \varphi(\hat{\theta}_n) \sim N(0,1)$ of this formula is about the distribution of the bootstrap replicates $\varphi(\hat{\theta}_n^*)$, with $\varphi(\hat{\theta}_n)$ an already observed, fixed value. Remember that on the right side of (8.14), or in (8.13), $\varphi(\hat{\theta}_n)$ is not a fixed value but a statistic, a random variable.

If the transform φ were known, (8.13) would give the exact $1-\alpha$ interval $\varphi(\theta) \in (\varphi(\hat{\theta}_n) + \Phi^{-1}(\frac{\alpha}{2}),\ \varphi(\hat{\theta}_n) + \Phi^{-1}(1-\frac{\alpha}{2}))$, and transforming it the exact $1-\alpha$ interval for θ:

$$\theta \in \left(\varphi^{-1}\{\varphi(\hat{\theta}_n) + \Phi^{-1}(\tfrac{\alpha}{2})\},\ \varphi^{-1}\{\varphi(\hat{\theta}_n) + \Phi^{-1}(1-\tfrac{\alpha}{2})\}\right) \tag{8.15}$$

(possible since the transform φ is monotone increasing).

In interval (8.15), $\varphi(\hat{\theta}_n) + \Phi^{-1}(\frac{\alpha}{2})$, expressing the lower limit, is a random variable on the right of (8.14) but a fixed value with respect to the left side. This fixed value is the expectation $\varphi(\hat{\theta}_n)$ of the bootstrap replicates

$$\varphi(\hat{\theta}_n^*) \sim N(\varphi(\hat{\theta}_n), 1)$$

shifted by $\Phi^{-1}(\frac{\alpha}{2})$. I.e. using the bootstrap approximation (8.14), $\varphi(\hat{\theta}_n) + \Phi^{-1}(\frac{\alpha}{2})$ is exactly the $\frac{\alpha}{2}$-quantile of the distribution $N(\varphi(\hat{\theta}_n), 1)$ of the random variable $\varphi(\hat{\theta}_n^*)$.[4] Likewise the upper limit $\varphi(\hat{\theta}_n) + \Phi^{-1}(1 - \frac{\alpha}{2})$ is approximately the $(1 - \frac{\alpha}{2})$-quantile of the distribution of $\varphi(\hat{\theta}_n^*)$.

For simplicity take $B = 1999$, $\alpha = 0.05$, and the bootstrap replicates arranged in order $\hat{\theta}^*_{(1)} \le \cdots \le \hat{\theta}^*_{(1999)}$. Since φ is monotone increasing,

$$\varphi(\hat{\theta}^*_{(1)}) \le \cdots \le \varphi(\hat{\theta}^*_{(1999)}), \tag{8.16}$$

and the $\frac{\alpha}{2}$-quantile of the distribution of $\varphi(\hat{\theta}_n^*)$ is $\varphi(\hat{\theta}^*_{(50)})$, the $(1 - \frac{\alpha}{2})$-quantile $\varphi(\hat{\theta}^*_{(1950)})$. So the interval (8.15) is approximated by the percentile interval:

$$\begin{aligned}\theta &\in \left(\varphi^{-1}\{\varphi(\hat{\theta}^*_{(50)})\},\ \varphi^{-1}\{\varphi(\hat{\theta}^*_{(1950)})\}\right)\\ &= (\hat{\theta}^*_{(50)},\ \hat{\theta}^*_{(1950)})\\ &= (Q_{G^*}(\tfrac{\alpha}{2}),\ Q_{G^*}(1 - \tfrac{\alpha}{2})).\end{aligned}$$

The transform φ does not affect the computation of the percentile interval, so it need not be known. The bootstrap percentile interval has the advantage of great simplicity and is used often.

8.5 Adjusted percentile intervals

Whether the percentile interval succeeds depends on the existence of a transform φ satisfying assumption (8.13), and in general such a transform cannot be said always to exist. Adjusted percentile intervals weaken this transform assumption so that approximation is possible over a wider range of populations. Among adjusted percentile intervals we look at the most representative, the bias-corrected and accelerated (BCa) interval, and its special case, the bias-corrected interval.

Suppose we have an estimator $\hat{\theta}_n$ of the parameter θ and its bootstrap replicates $\{\hat{\theta}_n^{*(b)} : b = 1, \ldots, B\}$ (or the bootstrap sampling distribution G^* of $\hat{\theta}_n$). In the percentile interval we assumed a monotone increasing transform φ with $\varphi(\hat{\theta}_n) - \varphi(\theta) \sim N(0, 1)$; but $\varphi(\hat{\theta}_n)$ may be a biased estimator of $\varphi(\theta)$, and the standard error $\mathrm{se}(\varphi(\hat{\theta}_n))$ may depend on the value of $\varphi(\theta)$. Taking all this into account, assume that for an unknown b (bias-correction factor), a (acceleration factor, or skewness-correction factor) and an unknown monotone increasing transform φ,

$$\varphi(\hat{\theta}_n) - \varphi(\theta) \sim N(-b\sigma_\varphi, \sigma_\varphi^2), \qquad \sigma_\varphi = 1 + a\varphi(\theta). \tag{8.17}$$

If $a = 0$, then $\mathrm{se}(\varphi(\hat{\theta}_n)) = \sigma_\varphi = 1$ for every value of $\varphi(\theta)$. If $b = 0$, $\varphi(\hat{\theta}_n)$ is an unbiased estimator of $\varphi(\theta)$. Why the unknown b is called the bias-correction factor is obvious, but why the unknown a is called the acceleration or skewness-correction factor rather than a "variance-correction factor" is hard to explain simply at the level of this book; just accept it for now. The form $\sigma_\varphi = 1 + a\varphi(\theta)$ of the standard error can be understood as a first-order approximation of a possibly more complicated standard error. As with the percentile interval, if a transform φ satisfying (8.17) merely exists, the

[4] In $\varphi(\hat{\theta}_n) - \varphi(\theta) \sim N(0, 1)$, $\varphi(\hat{\theta}_n)$ is a random variable and $\varphi(\theta)$ a fixed parameter. So the limits of interval (8.15) are random variables under $\varphi(\hat{\theta}_n) - \varphi(\theta) \sim N(0, 1)$, while under $\varphi(\hat{\theta}_n^*) - \varphi(\hat{\theta}_n) \sim N(0, 1)$ (the left side of (8.14)) they are fixed values. A fixed value $\varphi(\hat{\theta}_n) + \Phi^{-1}(p)$ is the p-quantile of the distribution $N(\varphi(\hat{\theta}_n), 1)$ of $\varphi(\hat{\theta}_n^*)$, which is not hard to see.

adjusted percentile interval can be obtained even without knowing φ. The two unknowns b and a must be estimated from the data, but first suppose b and a are known.

The model (8.17) gives the exact $1-\alpha$ interval for the parameter $\varphi(\theta)$

$$\varphi(\theta) \in \left(\frac{\varphi(\hat{\theta}_n) + b + \Phi^{-1}(\frac{\alpha}{2})}{1 - a\{b + \Phi^{-1}(\frac{\alpha}{2})\}},\ \frac{\varphi(\hat{\theta}_n) + b + \Phi^{-1}(1-\frac{\alpha}{2})}{1 - a\{b + \Phi^{-1}(1-\frac{\alpha}{2})\}} \right) := (L, U). \tag{8.18}$$

Coverage of (8.18) (supplied; Exercise 8.3 of the book). Write $\hat{\varphi} = \varphi(\hat{\theta}_n)$, $\phi = \varphi(\theta)$, and assume $1 + a\phi > 0$. By (8.17), $Z := \frac{\hat{\varphi} - \phi}{1 + a\phi} + b \sim N(0,1)$. Solving for ϕ,

$$\phi = g(Z) := \frac{\hat{\varphi} - (Z - b)}{1 + a(Z - b)}, \qquad g'(Z) = -\frac{1 + a\hat{\varphi}}{(1 + a(Z-b))^2} < 0$$

(when $1 + a\hat{\varphi} > 0$). So g is decreasing, and with $z_l = \Phi^{-1}(\frac{\alpha}{2})$, $z_u = \Phi^{-1}(1-\frac{\alpha}{2}) = -z_l$,

$$g(-z_l) = \frac{\hat{\varphi} + b + z_l}{1 - a(b + z_l)} = L, \qquad g(-z_u) = \frac{\hat{\varphi} + b + z_u}{1 - a(b + z_u)} = U.$$

Hence $\phi \in [L, U] \iff Z \in [-z_u, -z_l] = [z_l, z_u]$, which has probability $1 - \alpha$. ■

Since the transform φ is monotone increasing, the interval for θ is obtained with the inverse transform of φ: $\theta \in (\varphi^{-1}(L), \varphi^{-1}(U))$ is the exact $1-\alpha$ interval for θ. We do not know φ, so this interval will be approximated by the bootstrap.

Rewriting (8.17) as a pivot following the standard normal:

$$\frac{\varphi(\hat{\theta}_n) - \varphi(\theta)}{1 + a\varphi(\theta)} + b \sim N(0,1).$$

As in the percentile bootstrap, if a transform φ satisfying (8.17) exists, the bootstrap-approximated pivot also (approximately) follows the standard normal:

$$\frac{\varphi(\hat{\theta}_n^*) - \varphi(\hat{\theta}_n)}{1 + a\varphi(\hat{\theta}_n)} + b \overset{d}{\approx} \frac{\varphi(\hat{\theta}_n) - \varphi(\theta)}{1 + a\varphi(\theta)} + b \sim N(0,1).$$

Looking at the left side of this formula (again: $\varphi(\hat{\theta}_n)$ is a fixed value on the left and a random variable on the right), the transformed bootstrap replicates $\varphi(\hat{\theta}_n^*)$ approximately follow the normal $N(\mu_*, \sigma_*^2)$ with

$$\mu_* = \varphi(\hat{\theta}_n) - b\{1 + a\varphi(\hat{\theta}_n)\}, \qquad \sigma_*^2 = \{1 + a\varphi(\hat{\theta}_n)\}^2.$$

The limits (L, U) of the interval (8.18) for $\varphi(\theta)$ are no longer random variables but fixed values, so they can be expressed by the quantile function Q^* of $\varphi(\hat{\theta}_n^*) \sim N(\mu_*, \sigma_*^2)$ (Figure 8.5 of the book). The interval (8.18) for $\varphi(\theta)$ is expressed by the p_l- and p_u-quantiles of the distribution of $\varphi(\hat{\theta}_n^*)$:

$$(L, U) = (Q^*(p_l),\ Q^*(p_u)). \tag{8.19}$$

By the same logic as in the percentile interval,

$$\theta \in (Q_{G^*}(p_l),\ Q_{G^*}(p_u))$$

is the bootstrap approximation of the exact interval $\theta \in (\varphi^{-1}(L), \varphi^{-1}(U))$. Here the two probabilities p_l, p_u, obtained by solving (8.19), are

$$\begin{aligned} p_l &= \Phi\Big(\frac{L - \mu_*}{\sigma_*}\Big) = \Phi\Big(\frac{b + \Phi^{-1}(\frac{\alpha}{2})}{1 - a(b + \Phi^{-1}(\frac{\alpha}{2}))} + b\Big), \\ p_u &= \Phi\Big(\frac{U - \mu_*}{\sigma_*}\Big) = \Phi\Big(\frac{b + \Phi^{-1}(1-\frac{\alpha}{2})}{1 - a(b + \Phi^{-1}(1-\frac{\alpha}{2}))} + b\Big). \end{aligned} \tag{8.20}$$

Derivation of (8.20) from (8.19) (supplied; Exercise 8.4 of the book). Write $\hat{\varphi} = \varphi(\hat{\theta}_n)$, $w = b + \Phi^{-1}(\frac{\alpha}{2})$. From (8.18), $L(1-aw) = \hat{\varphi} + w$, so $L - \hat{\varphi} = w + aLw = w(1 + a\frac{\hat{\varphi}+w}{1-aw}) = \frac{w(1+a\hat{\varphi})}{1-aw}$. Hence

$$\frac{L-\mu_*}{\sigma_*} = \frac{L - \hat{\varphi} + b(1+a\hat{\varphi})}{1+a\hat{\varphi}} = \frac{w}{1-aw} + b,$$

and $p_l = P(\varphi(\hat{\theta}_n^*) \le L) = \Phi((L-\mu_*)/\sigma_*)$ is as stated. The same computation with $\Phi^{-1}(1-\frac{\alpha}{2})$ gives p_u. ■

Substituting the two probabilities (8.20) into (8.19), the $1-\alpha$ **bias-corrected and accelerated (BCa) percentile interval** is

$$\theta \in \left(Q_{G^*}\Big[\Phi\Big\{\frac{b + \Phi^{-1}(\frac{\alpha}{2})}{1 - a(b + \Phi^{-1}(\frac{\alpha}{2}))} + b\Big\}\Big],\ Q_{G^*}\Big[\Phi\Big\{\frac{b + \Phi^{-1}(1-\frac{\alpha}{2})}{1 - a(b + \Phi^{-1}(1-\frac{\alpha}{2}))} + b\Big\}\Big] \right).$$

If $a = 0$ (or one assumes so), a simpler expression is obtained; the interval

$$\theta \in (Q_{G^*}[\Phi\{2b + \Phi^{-1}(\tfrac{\alpha}{2})\}],\ Q_{G^*}[\Phi\{2b + \Phi^{-1}(1-\tfrac{\alpha}{2})\}])$$

contains only the bias-correction factor b, so it is called the **bias-corrected (BC) percentile interval**.

8.5.1 Estimating the bias-correction and acceleration factors

As in the percentile interval, the transform φ satisfying (8.17) need not be known, but the bias-correction factor b and acceleration factor a are estimated from the data as follows.

Bias-correction factor: $\hat{b} = \Phi^{-1}(G^*(\hat{\theta}_n)) = \Phi^{-1}(\#\{\hat{\theta}_n^{*(b)} \le \hat{\theta}_n\}/(B+1))$. If $\hat{\theta}_n$ is the median of the bootstrap sampling distribution, $\hat{b} = 0$ (then it is called median-unbiased).

Why this estimates b (supplied)

Under (8.17) and its bootstrap version, $P(\varphi(\hat{\theta}_n^*) \le \varphi(\hat{\theta}_n)) = \Phi((\hat{\varphi} - \mu_*)/\sigma_*) = \Phi(b)$. Since φ is increasing, the left side is $P(\hat{\theta}_n^* \le \hat{\theta}_n) = G^*(\hat{\theta}_n)$. So $b = \Phi^{-1}(G^*(\hat{\theta}_n))$.

Acceleration factor: to estimate the acceleration or skewness-correction factor a, a resampling independent of the bootstrap resampling is carried out. Recall that the statistic $\hat{\theta}_n$ is computed from all n observations, $\hat{\theta}_n = T(\mathbb{X}_n)$. The resampling used here deletes the observations one at a time and is repeated n times. With $\mathbb{X}_{(i)} := \{X_1, \dots, X_{i-1}, X_{i+1}, \dots, X_n\}$ the data without the ith observation X_i, the recomputed statistic $\hat{\theta}_{(i)} = T(\mathbb{X}_{(i)})$, and their mean $\hat{\theta}_{(\cdot)} = \frac{1}{n}\sum_{i=1}^n \hat{\theta}_{(i)}$,

$$\hat{a} = \frac{1}{6} \frac{\sum_{i=1}^n \left(\hat{\theta}_{(\cdot)} - \hat{\theta}_{(i)}\right)^3}{\left[\sum_{i=1}^n \left(\hat{\theta}_{(\cdot)} - \hat{\theta}_{(i)}\right)^2\right]^{3/2}} \tag{8.21}$$

estimates the acceleration factor a. This expression has the form of an estimator of skewness divided by 6, which hints why it is called the "skewness-correction factor."

The resampling used to estimate the acceleration factor is called the **jackknife** (Quenouille, 1949). The name jackknife (a pocket knife) was later used by Tukey (1958) when he employed it to estimate the bias and standard error of estimators. The bootstrap method can be seen as an extension of the jackknife.

Example 8.6 (Bias-corrected percentile interval)

Using nonparametric bootstrap resampling for $\hat{\theta}_n = \bar{X}_n$ of the Team Fortress 2 data, compute the 95% bias-corrected percentile interval for the population mean θ. First, to obtain the bias-correction factor $\hat{b}$, comparing the bootstrap sampling distribution or replicates $\hat{\theta}_n^{*(b)}$ with $\hat{\theta}_n$ gives $\#\{\hat{\theta}_n^{*(b)} \leq \hat{\theta}_n\}/(B+1) = 0.509$: the median of this distribution is (slightly) smaller than $\hat{\theta}_n$, a downward bias.

```
> hat.theta = mean(TFtime)
> n = length(TFtime); B = 1999
> theta.boot = replicate(B, mean(sample(TFtime, n, replace =
    TRUE)))
> b.factor = qnorm(sum(theta.boot <= hat.theta)/(B+1))
> b.factor
[1] 0.02381522
```

To correct the bias (towards smaller values) of the estimate, the lower and upper limits are adjusted slightly "upward."

```
> (BC.int.probs = pnorm(2*b.factor + c(qnorm(0.025), qnorm
    (1-0.025))))
[1] 0.02791671 0.97765680
```

Where the percentile interval used the 2.5% and 97.5% percentiles, the bias-corrected interval uses the 2.8% and 97.8% percentiles.

```
> (BC.interval = quantile(theta.boot, BC.int.probs))
2.791671% 97.76568%
 60.96237  80.75818
```

Example 8.7 (Bias-corrected and accelerated percentile interval)

Next, the 95% BCa percentile interval for θ. The bias-correction factor was already computed, so carry out the jackknife resampling (8.21) for the acceleration factor $\hat{a}$.

```
> theta.jack = vector(length = n)
> for (i in 1:n){theta.jack[i] = mean(TFtime[-i])}
> acc = sum((mean(theta.jack) - theta.jack)^3)/
+   (6 * sum(( mean(theta.jack) - theta.jack)^2)^(3/2))
> acc
[1] 0.02325921
```

Compared with the bias-corrected interval, the interval is adjusted further "upward."

```
> bz = b.factor + c(qnorm(0.025), qnorm(1-0.025))
> (BCa.int.probs = pnorm( (bz)/(1 - acc * (bz)) + b.factor ))
[1] 0.03370732 0.98229140
> (BCa.interval = quantile(theta.boot, BCa.int.probs))
3.370732% 98.22914%
 61.34667  81.40983
```

8.6 Comparing confidence intervals

We have looked at several ways of resampling the estimator $\hat{\theta}_n$ by the bootstrap to obtain intervals for the parameter θ. In particular, with the "Team Fortress 2" data, observations from a population

with a very heavy right tail, we computed intervals for the population mean θ using $\hat{\theta}_n = \bar{X}_n$. The table summarizes the seven 95% intervals; "BC" means the bias-corrected and "BCa" the bias-corrected and accelerated percentile interval.

	lower	upper
normal-approximation interval	60.95384	80.33238
normal-approximation bootstrap interval	60.91453	80.52683
basic bootstrap interval	60.78312	80.57468
bootstrap-t interval	**61.82523**	**82.17272**
percentile interval	60.71154	80.50310
BC percentile interval	60.96237	80.75818
BCa percentile interval	**61.34667**	**81.40983**

The sample is large enough here, so the normal approximation by the CLT is somewhat accurate and the normal-approximation interval

$$\bar{X}_n \pm z_{\alpha/2} \cdot \frac{s}{\sqrt{n}} = (60.9, 80.3) \tag{8.22}$$

is also somewhat accurate. Why use bootstrap intervals anyway? The answer has two parts.

The first is when the standard error or sampling distribution of the estimator cannot be derived: when the asymptotic distribution of the estimator is unknown, or it cannot be asserted to converge to a normal, or it converges to a normal but the sample is not large enough to use the normal approximation. Then a bootstrap interval, approximating the unknown sampling distribution from the data, is a good answer.

The second is the accuracy of intervals. When the population distribution cannot be specified, the coverage probability $P(\theta \in \text{CI})$ of all intervals introduced so far cannot be asserted to equal the given level $1 - \alpha$. But it is known that as the sample grows, the coverage converges to the given level. If the population is not normal but the estimator or pivot converges to a normal, the coverage of a normal interval like (8.22) is known to be

$$P(\theta \in \bar{X}_n \pm z_{\alpha/2} s/\sqrt{n}) = 1 - \alpha + O(n^{-1/2}).^{5}$$

That is, the difference between the coverage probability and the level converges to 0 at the rate $n^{-1/2}$. Among bootstrap intervals, the bootstrap-t and the bias-corrected and accelerated (BCa) percentile intervals are known to converge to the level faster:

$$\begin{aligned} P(\theta \in \text{"Bootstrap-}t\text{ CI"}) &= 1 - \alpha + O(n^{-1}), \\ P(\theta \in \text{"BCa CI"}) &= 1 - \alpha + O(n^{-1}). \end{aligned}$$

Being more accurate than the normal interval, they are said to have "second-order accuracy." By comparison the normal-approximation bootstrap interval (when the normal approximation holds), the basic bootstrap interval, the percentile interval and the bias-corrected interval have difference $O(n^{-1/2})$ between coverage and level, "first-order accuracy." In the table above, the second-order accurate bootstrap-t and BCa intervals are similar to each other and differ from the other, first-order accurate intervals.

For the basic and bootstrap-t intervals, whether the "pivot" holds is very important. So one needs to find a suitable transform, build the interval with an approximate pivot, and convert it back to an interval for the original parameter. Also, when the standard error is hard to compute, the

[5] A sequence a_n is $O(n^{-1/2})$ if for some $0 < M < \infty$ and all n, $\frac{|a_n|}{n^{-1/2}} = n^{1/2}|a_n| \le M$: a_n converges to 0 as fast as $n^{-1/2}$ or faster. E.g. comparing $a_n = n^{-1/2}$ and $b_n = n^{-1}$, b_n converges to 0 faster than a_n; then $a_n = O(n^{-1/2})$, $b_n = O(n^{-1})$.

bootstrap-t interval requires second-level bootstrap resampling, so the computation time becomes very long.

Percentile methods are more convenient since they need no effort to find a transform to a pivot. Among them, the BCa percentile interval, which guarantees second-order accuracy, is widely recommended. However, when bootstrap sampling does not take place within one population (several populations, regression, etc.), there is no guarantee that the estimate of the acceleration factor by (8.21) is accurate.